L ETA PRESERVES THE RANK OF FINITE FREE COHOMOLOGY SHEAVES

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Cohomology of L eta. Cohomology of Sheaves, Lemma cohomology-lemma-eta-first-property. H^i(Leta_I M) =~ I^ otimes i otimes (H^i(M)/H^i(M)[I]).

Source reference. Original reference 0GT4: sources/stacks-chow/tags.txt, source lines 20954–20954. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0F8N: sources/stacks-chow/tags.txt, source lines 19048–19048. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Situation 0GT4 let M be an object of $D(\mathcal{O}_X)$. Let $x \in X$ with $\mathcal{O}_{X,x}$ nonzero. If $H^i(M)_x$ is finite free over $\mathcal{O}_{X,x}$, then $H^i(L\eta_{\mathcal{I}}M)_x$ is finite free over $\mathcal{O}_{X,x}$ of the same rank.*

Proof. Namely, say $f \in \mathcal{O}_{X,x}$ generates the stalk $\mathcal{I}_x$. Then f is a nonzerodivisor in $\mathcal{O}_{X,x}$ and hence $H^i(M)_x[f] = 0$. Thus by Lemma 0F8N we see that $H^i(L\eta_{\mathcal{I}}M)_x$ is isomorphic to $\mathcal{I}_x^i \otimes_{\mathcal{O}_{X,x}} H^i(M)_x$ which is free of the same rank as desired. □

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